

[T]-Theory: Music & Arts

P8 · Clinical Soma · For: The Musicologist

Aesthetic Field Perturbation — BRECVEMA as the physics of musical emotion.

BRECVEMA as a field basis:

- Brainstem reflex — startle, tension release
- Rhythmic entrainment — pulse, groove, tempo
- Emotional contagion — performer → listener field coupling
- Cognitive appraisal — genre expectations, narrative

Upper mechanisms:

- Visual imagery — music-induced inner film
- Episodic memory — autobiographical recall trigger
- Musical expectancy — syntactic violation response
- AJ Aesthetic judgement — integrated somatic response

I · The Aesthetic Field Perturbation

A musical stimulus $S(t)$ perturbs the 8D BRECVEMA field Ψ :

$$\dot{\Psi} = -\nabla H(\Psi) + J_S(t) + \eta(t)$$

where $J_S(t) = \sum_{\alpha} J_{\alpha}(t) \hat{e}_{\alpha}$ projects the stimulus onto each BRECVEMA mechanism basis vector. The emotional response is the trajectory $\Psi(t)$ toward the nearest attractor basin.

Aesthetic judgement (AJ) is the gauge mode: $\sum_j W_{7j} = 0$ (G_2 zero-sum hypothesis). AJ measures the integrated field — it doesn't drive dynamics, it evaluates the result.

II · Why Music Is Universal

Music universality $\equiv$ the BRECVEMA basis is biologically fixed. The mechanisms are neurochemically anchored:

- **RE** rhythm: dopamine (nucleus accumbens) $\leftrightarrow$ motor loop
- **EM** memory: hippocampal reactivation via musical cue
- **ME** expectancy: prediction error signal in auditory cortex
- Chills = field crossing threshold T_c momentarily

Cultural variation = different attractor landscapes, same propagator.

III · The Green's Function of Music

$$G_{\text{music}}(\tau) = e^{-|\tau|/\tau_m} \cos(\omega_0 \tau)$$

A **damped oscillator**: the memory of a musical phrase decays exponentially with a superimposed oscillation at the tonic frequency ω_0 . This is identically the Green's function of a driven harmonic oscillator — the SHO = G identity at the musical scale (P11).

IV · Experimental Protocol

- Record physiological markers (HRV, skin conductance, EEG) during BRECVEMA-coded stimuli
- Fit τ_m from exponential envelope of emotional response
- Extract W_8 coupling matrix from cross-BRECVEMA correlations
- Test: $\sum_j W_{7j} \approx 0$ for AJ (gauge hypothesis)

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